No Genuine Copula in Japanese: a reanalysis of 'da' as a conceptual-locative and conjunctive postposition combined with a one-argument existential verb[1]

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Abstract

This paper proposes an alternative analysis of the Japanese so-called copula *da* and its non-contracted form *dearu*. Traditionally, *da/dearu* has been treated as a copula equivalent to English *to be*. However, the morphosyntactic and prosodic separability of *de* and *aru* in *dearu*, together with the syntactic and prosodic constituency of *de* and the preceding noun phrases, suggests that this interpretation is inadequate. Instead, we propose that *da/dearu* should be reanalysed as **the conceptual-locative and conjunctive postposition =*de* combined with the one-argument existential verb *aru***.

Under this analysis, *A=wa B=de aru* 'A is B' is syntactically both an **existence-asserting** and a **domain-specifying** sentence. It consists of the **ONE-ARGUMENT core construction [*A=wa aru*] 'A exists'** and the **ADJUNCT [*B=de*] 'and it is in the conceptual domain of B'** or, more simply, **'and it is as B'**. Schematically, this structure can be represented as the **conjunctive proposition EXIST(A) ∧ IN(A, B)**, distinct from the atomic BE(A, B) encoded by *to be*. Although E(A) ∧ IN(A, B) is logically redundant, it is taken here to be indispensable for the formalisation of linguistic form. The syntactically adjunctive *B=de* is nevertheless non-deletable owing to its pragmatic and informational indispensability, as it functions as the **focus of the sentence**, whereas the syntactically obligatory *kare=wa aru* represents the **presuppositional background**.

Based on this reanalysis, it becomes possible to offer a unified account of the structure, semantics and historical development of Japanese *da/dearu* constructions.

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1. Introduction — morphosyntactic and prosodic separability of *de* and *aru* in *dearu* and syntactic and prosodic constituency of *de* and the preceding NPs

In Japanese linguistics, *da/dearu* has traditionally been treated as a copula equivalent to *to be* in English, as in (1), where *dearu* can be replaced by =*da*, '=' indicating enclisis:

(1)	kare=wa	gakusei	dearu
	he(SUBJ)=TOP	student(PRED)	COP
	'He is a student.'		

However, this analysis faces a major challenge: it cannot account for the **high degree of independence between *de* and *aru* in *dearu***. For instance, when the noun phrase (NP) *gakusei* 'student' is marked by the enclitic focus/additive particle =*mo* 'also' — one of the so-called *toritatesi* in Japanese linguistics — the sentence cannot take the expected form shown in (2):

(2)	*kare=wa	gakusei	=mo	**dearu**
	he(SUBJ)=TOP	student(PRED)	also(ADD)	COP
	'He is also a student.'			

but must be as in (3):

(3)	kare=wa	gakusei	**de**	=mo	**aru**
	he(SUBJ)=TOP	student(PRED)	COP?	also(ADD)	COP?
	'He is also a student.'				

The particle =*mo* is inserted between *de* and *aru*, as if it were an infix — a situation comparable to the adverb *also* being inserted into the copula *is* in English, as in **he i- also -s a student*. Furthermore, in (4):

(4)	kare=wa	gakusei	**deari**	kanozyo=wa	syakaizin	**=da**
	he(SUBJ)=TOP	student(PRED)	COP	she(SUBJ)=TOP	working adult(PRED)	COP
	'He is a student, she is a working adult.'					

the first *deari* may be reduced to *de* alone, without *ari* — formally, the conjunctive form, or *ren'yōkei*, of the verb *aru* — as in (5):

(5)	kare=wa	gakusei	**de**	kanozyo=wa	syakaizin	**=da**
	he(SUBJ)=TOP	student(PRED)	COP?	she(SUBJ)=TOP	working adult(PRED)	COP
	'He is a student, she is a working adult.'					

Prosodically, too, the so-called copula *dearu* exhibits unexpected behaviour. If *dearu* were a single word, it would be expected to constitute either an independent prosodic word (ω) or an enclitic, so the utterance *kare=wa gakusei dearu* would be prosodised as either (6) or (7), where " | " marks a prosodic-word boundary, and prosodic words in Japanese generally correspond to traditional *bunsetu*:

(6) | *kare=wa* | *gakusei***=*dearu*** | — *dearu* as an enclitic (i.e. encliticised to *gakusei*)

(7) | *kare=wa* | *gakusei* | ***dearu*** | — *dearu* as an independent ω

In actual speech, nevertheless, the prosodisation is implemented as in (8):

(8) | *kare=wa* | *gakusei=****de* | *aru*** | — *de* as an enclitic and *aru* as an independent ω

Since prosodic-word boundaries are determined with reference to morphosyntactic information (Nespor & Vogel 2007), the fact that the utterance is prosodised as | *kare=wa* | *gakusei=de* | *aru* | strongly suggests the presence of a morphosyntactic boundary between *de* and *aru* — namely, that *de* and *aru* constitute distinct morphosyntactic elements. This prosodic separation is further supported by the fact that the discourse particle =*ne* can intervene between *de* and *aru*, which would be impossible if *dearu* constituted a single prosodic word.

All of these facts demonstrate that ***de* and *aru* are morphosyntactically and prosodically separable**, whereas **the NP *gakusei* and *de* are always syntactically and prosodically inseparable**, as illustrated in (9):

(9) *Kare=wa* [*gakusei de*] [*aru*].

Kare=wa [*gakusei de*] *mo* [*aru*].

Kare=wa [*gakusei de*] [*ari*], *kanozyo=wa syakaizin=da.*

Kare=wa [*gakusei de*], *kanozyo=wa syakaizin=da.*

Hence, in *kare=wa gakusei dearu*, the sequence *gakusei de* is assumed to constitute a syntactically and prosodically defined constituent, as Dalla Chiesa (2015) notes: "the NP-*de* element must always be the phrase (in the closest proximity to the verb)"[2], where the parenthetical remark is by the present author. This *gakusei de* is formally identical to a **postpositional phrase (PP) consisting of a NP combined with the postposition =*de***. In what follows, hence, *gakusei de* will be treated as a PP, that is, the PP *gakusei=de*. This separation between =*de* and *aru* is compatible with Matsuoka's observation that "the copula *da* consists of the postposition *de* and the verb *aru*" (2019: 142). The fact that *gakusei=de* and *aru* each form separate prosodic words can thus be naturally accounted for.

2. NP + =*de* as a conceptual-locative adjunct PP combined with the one-argument existential verb *aru*

In Japanese, **PPs of the form [NP + =*de*] generally function as adjuncts**, as Hasegawa (1999: 26, translated from Japanese) states: "The case particle =*de* occurs not only for 'location of an action' but also for a variety of other elements such as 'cause' (*kaze=de yasumu* 'to be absent because of a cold'), 'means' (*genkin=de kau* 'to buy with cash'), 'time' (*mikka=de tukuru* 'to make in three days') and 'state' (*hadaka=de odoru* 'to dance naked'), and all of these function fundamentally only as adjuncts. There are no predicates that require these as arguments indispensable for informational completeness. From this, one can generalise that =*de* marks adjuncts." Based on this generalisation, *kare=wa gakusei=de aru* can be syntactically analysed as consisting of the **obligatory structure [*kare=wa aru*]** and the **adjunctive structure [*gakusei=de*]**. Further justification for treating *gakusei=de* not as an argument but rather as an adjunct will be provided below in this section and in Section 7.

In *kare=wa aru*, the predicate *aru* — or, more precisely, its stem *ar-* — can thus be analysed as a **ONE-ARGUMENT verb**, taking only the NP *kare* as its argument. When the verb ***aru* functions as one-argument predicate**, it means '**to exist**' — though in a rather literary sense — as in *ware omou, yueni ware* ***ari*** 'I think, therefore I am', and "***aru***"*=to iu koto=wa dō iu koto=ka=to iu toi*

[2] She argues that the ungrammaticality of **neko=**de** wagahai=ga* ***aru*** 'I am a cat' supports her claim that "the NP-*de* element must always be the phrase in the closest proximity to the verb". However, the ungrammaticality of this sentence does not arise from word order, but rather from the fact that *wagahai*, while functioning as a topic, is not marked by the particle =*wa*. Indeed, although rather awkward, *neko=**de** wagahai=**wa** **aru*** is not completely ungrammatical, and expressions such as NP=***de*** *watasi=**wa** **ari**-tai* 'as the referent denoted by the NP I want to exist' — e.g. *ryōsikitekina syakaizin=**de** watasi=**wa** **ari**-tai* 'as a rational member of society I want to exist' — sound much more natural.

'the problem of what it means "to exist"'. The sentence *kare=wa aru* can thus be interpreted as 'he exists', **asserting the existence** of *kare*. This meaning is reflected in the current everyday compound nouns ***ari**kata* and ***ari**yō*, which signify 'mode/manner of existing, ideal way to exist' and 'state/appearance of existing', respectively. Both are derived from the stem *ari-* of the verb *aru* in this sense — namely, 'to exist' — combined with the nominal formatives *-kata* 'mode, manner, way' and *-yō* 'state, appearance', respectively. Further justification for analysing *aru* in this construction as 'to exist' — a meaning that, for pragmatic and informational reasons, is rarely brought to the speaker's or hearer's awareness in ordinary categorical or identificational contexts of the type 'A = B' or, more precisely, 'A ⊆ B' — will be presented in Section 5.

As for the PP *gakusei=de*, on the other hand, although **PPs headed by *=de*** can denote, as mentioned above, a variety of semantic roles such as location, cause, means, time, or state, the most plausible interpretation in this construction is **locative** — namely, its primitive sense — though not denoting a physical, but a **conceptual, location**. Conceptual location here refers to the localisation or anchoring of an entity within a conceptual space structured by domains mutually defined through semiological opposition, founded on an analogy with physical space. Consequently, *gakusei=de* means 'in the **conceptual domain** of "student"' or, more simply, 'as a student'.

The entire structure *kare=wa gakusei=de aru* is thus, syntactically, composed of the **ONE-ARGUMENT obligatory construction [*kare wa aru*]** 'he exists' and the **ADJUNCT [*gakusei=de*]** 'and in the conceptual domain of "student"' or 'and as a student'. Semantically, it expresses 'He exists, and his existence is located in the conceptual domain of "student"' or, more simply, 'He exists, and it is as a student'. Schematically, *A=wa B=de aru* — in which *A=wa aru* 'A exists', even without *B=de* 'in the conceptual domain of B', constitutes an atomic proposition, since *B=de* is assumed to be adjunctive — can be represented as **E(A) ∧ IN(A, B)**, where E(A) 'A exists' — E abbreviates EXIST — stands for **the assertion of existence**, and IN(A, B) 'A is in B' for **the specification of the domain of existence**. E(A) ∧ IN(A, B) thus constitutes a **conjunctive proposition**. Note that, since IN(A, B) ⇒ E(A), E(A) ∧ IN(A, B) ≡ IN(A, B); thus, E(A) ∧ IN(A, B) is logically redundant, although it is still assumed to be indispensable in the formalisation of linguistic form.

The remaining problem is to determine what element in the linguistic form *A=wa B=de aru* formally corresponds to the operator ∧, which expresses logical conjunction in the proposition E(A) ∧ IN(A, B). As will be discussed in Section 6, the contemporary postposition *=de* originated historically from the sequence =*ni-te*, where =*ni* functioned as a locative postposition and -*te* as a conjunctive suffix. The development was probably driven by the deletion of -*i*- and by postnasal voicing — typical of the Yamato lexical stratum — yielding the sequence =*ni-te* → =*n-te* → =*n-de* → =*de*. Even today, in PPs of the form [NP + =*de*], one can observe stylistic variation between the postposition =*de* and the postposition-suffix sequence =*ni-te*, as in *Tōkyō*=***de*** / *Tōkyō*=***ni-te*** 'in Tokyo, at Tokyo', though the latter is a more formal and literal form. Given this diachronic development and the synchronic variation, it is plausible to assume that the underlying form of *B*=*de* is still *B*=*ni-te*, where, strictly speaking, ***B*=*ni* denotes conceptual location** and **the conjunctive suffix -*te* adjoins *B*=*ni*, not as an argument but as an adjunct, to *A*=*wa aru*** — or more precisely, to the VP node within *A*=*wa aru*.

An NP marked by =*ni* functioning as an argument is required by the predicate as an element semantically indispensable for completing the predicate's meaning. The link between the predicate and such an argumental *NP*=*ni* is determined by the argument structure inherent in the predicate, and is therefore intrinsic to it. By contrast, an underlying *NP*=*ni* functioning as an adjunct is optional for the predicate or for some node in the predicate phrase, and the link between the adjunct *NP*=*ni* and the predicate or such a node is extrinsic to the predicate. In Contemporary Japanese, this extrinsic relation needs to be made explicit in the linguistic form, and the device used for this purpose is the conjunctive suffix -*te*. It may be assumed, from a cognitive perspective, that the *conjunctivity* of -*te* has been metaphorically extended — via the schema of *junctivity* — into *adjunctivity*, as illustrated in (10):

junctivity

(10) metaphor: **conjunctivity** -------> **adjunctivity**

Having thus developed into an adjunctive suffix, -*te* adjoins *NP*=*ni* to the predicate or to some node in the predicate phrase as an adjunct rather than as an argument. The surface realisation of this *NP*=***ni-te*** is *NP*=***de***. Thus, the use of =*de* as an adjunct marker in Contemporary Japanese can be accounted for by the fact that its underlying form =*ni-te* contains the conjunctive suffix -*te*, which functions adjunctively in this context. Consequently, in the construction *A*=*wa B*=***de*** *aru*, it seems accurate to characterise =*de* as a **conceptual-locative and conjunctive postposition**.

In opposition to this *A*=*wa* ***B*=*de*** *aru*, the semantically related and syntactically similar construction *A=wa* ***B=ni*** *aru/iru* 'A is located in B' — in which *A=wa aru/iru* 'A is located', without *B=ni* 'in B', cannot constitute a proposition, since *B=ni* is argumentative, as dicussed in Section 3 — can be represented as **IN(A, B)** 'A is located in B', which denotes the **specification of the domain of location**. IN(A, B) thus constitutes an **atomic proposition**. The compositional difference between the conjunctive E(A) ∧ IN(A, B) and the atomic IN(A, B) is reflected in their default negation patterns, such as the partial negation *A=wa* <u>*B=de=**wa***</u> ***nai*** vs. the total negation *A=wa* <u>*B=ni*</u> ***nai/i-nai*** — namely, E(A) ∧ ¬IN(A, B) vs. ¬IN(A, B) — which will be discussed in Section 7.

Thus, instead of the traditional **copula-based** analyis in (11) = (1), representable as BE(A, B):

(11) kare=wa gakusei **dear**-u
he(SUBJ)=TOP student(PRED) **COP**-PRES
'He is a student.'

in this paper, a **conceptual-locative/conjunctive-postposition-plus-one-argument-existential-verb-based** analysis in (12), representable as E(A) ∧ IN(A, B), is adopted, where CONCEP-LOC and CONJ abbreviate 'conceptual-locative' and 'conjunctive', respectively:

(12) kare=wa gakusei=**de** **ar**-u
he(SUBJ)=TOP student=**CONCEP-LOC/CONJ (ADJUNCT)** **exist**-PRES
'He exists, and his existence is in the conceptual domain of "student".'

3. The differences between *sore wa gakkō=DE ARu* and *sore wa gakkō=NI ARu*

In contrast to the construction *sore=wa gakkō=**de** **aru*** in (13), hypothesised as **existence-asserting and domain-specifying** in Section 2:

(13) sore=wa gakkō=**de** **ar**-u
it(SUBJ)=TOP school=**CONCEP-LOC/CONJ(ADJUNCT)** **exist**-PRES
'It exists, and its existence is located in the conceptual domain of "school".'

in genuine locative — namely, **location-specifying-only** — constructions as in (14) and (15), where PHYSIC-LOC abbreviates 'physical-locative':

(14) sore=wa gakkō=**ni** **ar**-u
it(SUBJ)=TOP school=**PHYSIC-LOC(ARGUMENT)** **be located**-PRES
'It is located in the physical domain of "school".'

(15)	kare=wa	gakkō=**ni**	**i**-ru
	he(SUBJ)=TOP	school= **PHYSIC-LOC(ARGUMENT)**	**be located**-PRES
	'He is located in the physical domain of "school".'		

the **physical-locative argument is obligatory**, as mentioned in Section 2, **and always marked by the postposition =*ni***, rather than =*de*. Thus, the verb *aru* — more precisely, its stem *ar-* — when used as a LOCATIVE **verb**, is a TWO-ARGUMENT **predicate with the argument structure [physical domain, theme]**, the location being denoted by the postpositional phrase [NP + =*ni*], as in *sore=wa gakkō=**ni aru***. When the theme is an animate entity, the verb *iru* with the stem *i-* is used instead as in (15), with the **argument PP [NP + =*ni*] unchanged**, as in *kare=wa gakkō=**ni iru*** 'he is in the school'. By contrast, the sequence of words in (16), in which the argumentative PP [NP + =*ni*] in (15) has been replaced by the PP [NP + =*de*]:

(16)	kare=wa	gakkō=**de**	**i**-ru
	he(SUBJ)=TOP	school=LOC/CONJ(**ADJUNCT**)	?-PRES
	' ? '		

is semantically uninterpretable — even for native speakers — and ungrammatical, indicating that the PP [NP + =*de*] is adjunctive in this construction.

Given the general adjunctivity of the PP [NP + =*de*] and its specific adjunctive function in purely locative constructions, *kare=wa gakusei=de aru* may reasonably be regarded SYNTACTICALLY — though NOT PRAGMATICALLY OR INFORMATIONALLY, as will be discussed in Section 4 — as a **structurally self-contained, existence-asserting construction** *kare=wa aru* 'he exists', **accompanied by the conceptual-locative adjunct** *gakusei=de* 'and it is in the conceptual domain of "student"', **which conjunctively specifies the domain of existence**.

Thus, in *A=wa B=de aru*, the constituent *B* denotes a conceptual domain of location, whereas in *A=wa B=ni aru/iru*, *B* denotes a physical domain of location. This difference surfaces even when B is replaced by a pro-form. As shown in (17) and (18), in the former construction *B* is substituted by the pro-form ***sō***, which expresses such notions as property, attribute or characteristic, glossable as "so"; in the latter construction, the element *B* is substituted by the locative demonstrative pronoun ***soko*** "there":

(17) *A=wa **B**=de aru.* → *A=wa **sō**=de aru.* (conceptual domain)

(18) *A=wa **B**=ni aru/iru* → *A=wa **soko**=ni aru/iru.* (physical domain)

Although both constructions involve a proposition of the form IN(A, B), the formal opposition between *sō* and *soko* reflects whether the second individual constant *B* denotes a conceptual domain or a physical domain. Note that even when the postposition is *=de*, if *B* denotes a physical domain, it is substituted by *soko*, as in *kare=wa* ***gakkō****=de hataraite=iru* 'he works / is working in the / a school → *kare=wa* ***soko****=de hataraite=iru* 'he works / is working there'.

4. The syntactic optionality and pragmatic necessity of *gakusei=de*

Although *gakusei=de* in *kare=wa gakusei=de aru* is **syntactically an adjunct**, it **cannot be deleted**. This is because the sentence, while structurally consisting of the obligatory core *kare=wa aru* 'he exists' and the adjunctive *gakusei=de* 'and in the conceptual domain of "student"' or 'and as a student', serves as a **categorical or identificational sentence** in which **the focus is on the adjunct *gakusei=de***.

Therefore, the **non-deletability of *gakusei=de* is assumed to arise from its pragmatic and informational indispensability** rather than from syntactic necessity. The distinction can be represented as in the following table (19):

(19)

Level	Structure	Function
Syntax	[$_{VP}$ *A ar-*] + [$_{PP}$ *B=de*]	One-argument VP + adjunct PP
Semantics	'A exists, and it is in the domain B'	Existence assertion + domain specification E(A) ∧ IN (A, B)
Information Structure	[*A=wa aru*] + [*B=de*]	Given + New (default)
Pragmatics	Focus on *B=de* (default)	Pragmatic obligatoriness and non-deletability of *B=de* due to focus

5. Shift of focus and foregrounding of the lexical meaning 'to exist' of the verb *aru*

In the construction *watasi=wa ryōsikitekina syakaizin=de aru* in (20):

(20) watasi=wa ryōsikitekina syakaizin=de **ar**-u
I(SUBJ)=TOP rational member of society=CONCEP-LOC/CONJ(ADJUNCT) **exist**-PRES
'I EXIST as a rational member of society.'

which can be schematically represented as **E(A) ∧ IN(A, B)**, the existence-asserting component E(A) — that is, the assertion of the lexical meaning 'to exist' encoded in the verbal stem *ar-* — is backgrounded as given information, whereas the domain-specifying component IN(A, B) is foregrounded as the focus of the utterance. As a result, the meaning 'to exist' is rarely brought to the speaker's or hearer's awareness in ordinary categorical or identificational contexts. By contrast, in example (21), which takes the form **A WANT [E(A) ∧ IN(A, B)]**, the situation is fairly different:

(21)	watasi=wa	ryōsikitekina syakaizin=de	**ari-**	-tai-0
	I(SUBJ)=TOP	rational member of society=CONCEP-LOC/CONJ(ADJUNCT)	**exist**	want-PRES
	'I want to EXIST as a rational member of society.'			

Here, what is at issue is 'one's mode of existence', that is, an orientation toward a particular manner of being. In such a context, **E(A) is no longer backgrounded but instead becomes part of the focus as well as IN (A, B)**, and **the lexical meaning 'to exist' encoded in *ar-* emerges as perceptually salient**. This pragmatic shift of focus, which foregrounds the existential meaning of *ar-*, supports empirically the validity of analysing **the semantic structure of *A=wa B=de aru* not as BE(A, B) but as EXIST(A) ∧ IN(A, B)**.

It should be noted that the English translation 'I want to BE a rational member of society', represented as **A WANT [BECOME(A, B)]** — based on 'I AM a rational member of society', represented as **BE(A, B)** — is a complete mistranslation of the construction *watasi=wa ryōsikitekina syakaizin=de ARI-tai* in (20). This in itself clearly demonstrates that the Japanese *=da/=de aru* is not equivalent to the English *to be*, nor does it function as a copula in the same way. The difference between them can be represented as in (22):

(22)	*Watasi=wa ryōsikitekina syakaizin=**de aru***:	**EXIST(A) ∧ IN(A, B)**
	*I **am** a rational member of society*:	**BE(A, B)**
	*Watasi=wa ryōsikitekina syakaizin=**de ari-tai***:	**A WANT [EXIST(A) ∧ IN(A, B)]**
	*I **want to be** a rational member of society*:	**A WANT [BECOME(A, B)]**

What corresponds in Japanese to the English *I want to be a rational member of society* in (22) is the sentence in (23):

(23) *Watasi=wa ryōsikitekina syakaizin=**ni nari-tai***: **A WANT [BECOME(A, B)]**

which can be represented as shown in (24), containing the two-argument state-change verb *naru* 'to become', with *=ni* marking the argument GOAL:

(24)	watasi=wa	ryōsikitekina syakaizin=ni	**nari-**	-tai-0
	I(SUBJ)=TOP	rational member of society=GOAL(ARGUMENT)	**become**	want-PRES
	'I want to BE a rational member of society.'			

6. From Old Japanese to Contemporary Japanese forms

In Old Japanese (OJ, 700–800), the modern formally distinct constructions *A=wa B=**de aru*** 'A exists (and it is) as B' and *A=wa B=**ni aru/iru*** 'A is (located) in B' would have corresponded to the seemingly identical construction *A B=**ni ari***. "Dative *ni* is the general oblique case, marking both argument and non-argument oblique nominals" (Frellesvig 2010: 126), unlike in Contemporary Japanese, where *=ni* marks arguments, while *=de* — which originated from the postposition-suffix sequence *=nite* — marks adjuncts, in accordance with the diachronic description "OJ *ni* and *nite* became more specialized with *ni* being used more for arguments and *nite* more for adjuncts; from mid to late EMJ *nite* acquired the variant *de* which is still in use in contemporary NJ" (*ibid.*: 243), where EMJ and NJ abbreviate Early Middle Japanese (800–1200) and 'new Japanese' — namely, Modern Japanese, 1600– — respectively.

Thus, the former *A B=ni ari* 'A exists (and it is) as B' can be analysed as an existence-asserting and domain-specifying construction consisting of the obligatory construction *A ari* 'A exists' — with the verb *ari* being used as **one-argument existential** predicate — and the conceptual-locative adjunct *B=ni* 'and in the conceptual domain referred to by B', as shown in (25). The latter *A B=ni ari* 'A is (located) in B', on the other hand, can be analysed as a location-specifying -only construction — in which *ari* is employed as a **two-argument locative** predicate — and *B=ni* 'in B' constitutes the physical-locative argument, as indicated in (26):

(25)	A	B=**ni**	**ari**-0
	A(SUBJ)=TOP	B=CONCEP-LOC(**ADJUNCT**)	**exist**-PRES
	'A exists (and it is) as B.'		

(26)	A	B=**ni**	**ari**-0
	A(SUBJ)=TOP	B=PHYSIC-LOC(**ARGUMENT**)	**be located**-PRES
	'A is located in B.'		

The Old Japanese constructions in (25) and (26) appear superficially identical, but when the element *B* is substituted by a pro-form, they diverge as in (27) and (28):

(27) *A **B**=ni ari.* → *A **sa**=ni ari.* (conceptual domain)

(28) *A **B**=ni ari.* → *A **soko**=ni ari.* (physical domain)

Consequently, although both constructions is assumed to involve a proposition of the form IN(A, B), the morphological opposition between *sa* and *soko* reflects whether the second individual constant *B* refers to a conceptual domain or to a physical domain.

These superficially identical constructions in (25) and (26), however, diverged diachronically into the formally distinct modern constructions in (29) and (30):

(29)	A=wa	B=**de**	**ar**-u
	A(SUBJ)=TOP	B=CONCEP-LOC/CONJ(**ADJUNCT**)	**exist**-PRES
	'A exists, and it is as B.'		

(30)	A=wa	B=**ni**	**ar**-u / **i**-ru
	A(SUBJ)=TOP	B=PHYSIC-LOC(**ARGUMENT**)	**be located**-PRES
	'A is located in B.'		

As already stated in Section 3, in Contemporary Japanese as well, the element *B* undergoes different pro-form substitutions, as in (31) and (32) = (17) and (18):

(31) *A=wa **B**=de aru.* → *A=wa **sō**=de aru.* (conceptual domain)

(32) *A=wa **B**=ni aru/iru* → *A=wa **soko**=ni aru/iru.* (physical domain)

In (30), the locative verbs *aru* and *iru*, both semantically corresponding to the Old Japanese *ari* — though the latter originally derived from another verb *wiru* 'to sit; to be seated' — are used for inanimate and animate entities, respectively. The historical development from (25) to (29) is as follows. In Old Japanese, there existed "analytic forms consisting of an infinitive (*ni* …) followed by *ar-*" (Frellesvig 2010: 95) for the so-called copula — that is, *ni ar-*. In the subsequent Early Middle Japanese (EMJ, 800–1200) period, this analytic form "*ni ar-* had the variant *nite ar-*" (*ibid.*: 235). "In the course of EMJ" (*ibid.*: 235), or more specifically "from the end of the period" (*ibid.*: 233), "*nite ar-* became *de ar-*" (*ibid.*: 235). As illustrated in "Table 12.8. *Development of the copula and adjectival-copula paradigms*" (*ibid.*: 344), from the end of EMJ to early Late Middle Japanese (early LMJ, Kamakura period 1185-1333), the three forms *ni ar-*, *nite ar-* and *de ar-* coexisted, although "already from late EMJ" (Insei period, 1086-1185), "*ni* was getting more restricted" and "*de* was used far more widely" than *nite*. The same table (*ibid.*: 344) shows that only the newest *de ar-* continued to be used from late LMJ (Muromachi period 1333-1573) into Contemporary Japanese.

The historical developments in (33) and (34):

(33) *A B=**ni** ari* → *A B=**ni-te** ari* → *A=wa B=**de** aru* — E(A) ∧ IN(A, B)

(34) *A B=**ni** ari* → *A B=**ni** ari* → *A=wa B=**ni** aru/iru* — IN(A, B)

may plausibly reflect the diachronic split between *=ni*, used for arguments, and *=ni-te* — later, *=de* — used for adjuncts, where the conjunctive suffix *-te* adjoins the PP [NP + *=ni*] as an adjunct, rather than as an argument, to another node, as discussed in Section 2. This suggests a shift from a linguistic stage in which *=ni* marked both arguments and adjuncts, without the help of the conjunction *-te*, to one in which *=ni* and *=ni-te/=de* became specialised as argumental and adjunctive markers, respectively. Thus, the historical split of the Old Japanese *=ni ari* between the Modern Japanese *=de aru* and *=ni aru/iru* may not reflect a reanalysis of a single construction, but rather the formal divergence of two distinct semantic structures — 'existence and its location in a conceptual domain' vs. 'location in a physical domain' — that happened to share the same surface form *A B=ni ari* in Old Japanese — a possibility that merits further historical investigation.

7. Default partial negation and the scope of negation marked by the particle =*wa*

The **default negation pattern of the existence-asserting and domain-specifying** sentence in (35) = (12):

(35)	kare=wa	gakusei=de	ar-u
	he(SUBJ)=TOP	student=CONCEP-LOC/CONJ(ADJUNCT)	exist-PRES
	'He exists, and it is as a student.'		

which, in line with Section 2, can be schematically represented as the conjunctive proposition E(A) ∧ IN(A, B), is the **partial negation** in (36):

(36) kare=wa gakusei=de=**wa** (*ara-) **-nai**-0
he(SUBJ)=TOP student=CONCEP-LOC/CONJ(ADJUNCT)=**SCOPE OF NEG** exist **NEG**-PRES
'He exists, but it is **not** as a student.'

rather than the total negation in (37):

(37) kare=wa gakusei=de (*ara-) **-nai**-0
he(SUBJ)=TOP student=CONCEP-LOC/CONJ(ADJUNCT) exist **NEG**-PRES
'He does **not** exists as a student.'

Note that in (36) and (37), the stem of the existential verb *aru* is represented as (*ara-), since in contemporary Standard Japanese the negative form of this verb is not the theoretically expected **ara-nai*, but obligatorily the suppletive adjective *nai*. The more natural sentence *kare=wa gakusei=de=**wa** **nai*** in (36) — or, more colloquially, *kare=wa gakusei=dy**a** **nai***, with the contracted form *=dya* derived from the sequence *=de=wa* — does **not negate the existence of *kare* itself**, but rather **negates only *gakusei=de* as the relevant domain of existence**. In this way, **the particle =*wa* in Japanese can mark the scope of negation**.

Schematically, the partial negation *A=wa B=de=**wa** **nai*** in (36) — underlyingly, **A=wa B=de=**wa** ara-**nai*** — can be represented as **$E(A) \wedge \neg IN(A, B)$**, whereas the total negation *A=wa B=de **nai*** in (37) — underlyingly, **A=wa B=de ara-**nai*** — can be represented as **$\neg(E(A) \wedge IN(A, B)) \equiv \neg E(A) \vee \neg IN(A, B)$**. Although this disjunction is compatible with $\neg E(A) \wedge IN(A, B)$, $E(A) \wedge \neg IN(A, B)$ or $\neg E(A) \wedge \neg IN(A,B)$, the first case is inconsistent with the conditional $IN(A, B) \Rightarrow E(A)$ — hence, the contrapositive $\neg E(A) \Rightarrow \neg IN(A, B)$ — and the third case, $\neg E(A) \wedge \neg IN(A, B) \equiv \neg E(A)$, is pragmatically infelicitous in topicalised constructions with *A=wa*, since *A=wa*, as a topicalised phrase, tends to presuppose the contextual or situational givenness — that is, the existence — of *A*, and since there are dedicated Japanese forms for existential negation such as *A=wa nai / i-nai / sonzai-si-nai*; in practice, therefore, the interpretation converges on $E(A) \wedge \neg IN(A, B)$. Hence, the former pattern in (36), which unambiguously denotes the partial negation $E(A) \wedge \neg IN(A, B)$, is judged to be the more natural one.

If *=de aru* were a single copula, then *A=wa B=de aru* would denote BE(A, B), and its negation would take the form of plain predicate negation — namely, *A=wa B=de nai*, representable as $\neg BE(A, B)$, with no *=wa* following *=de*. **Empirically, however, *A=wa B=de nai* is perceived as infelicitous, whereas *A=wa B=de=wa nai* is preferred**. This spontaneous preference for the particle *=wa* after *=de* is unexpected under the simple-copula analysis, but follows naturally if the sentence is decomposed as $E(A) \wedge IN(A, B)$, where the logically and pragmatically most plausible interpretation is the partial negation E(A) ∧ ¬IN(A, B).

By contrast, the **default negation of the location-specifying-only** (38) and (39) = (14) and (15):

(38)	sore=wa	gakkō=ni	ar-u
	it(SUBJ)=TOP	school=PHYSIC-LOC(ARGUMENT)	be located-PRES
	'It is located in the school.'		

(39)	kare=wa	gakkō=ni	i-ru
	he(SUBJ)=TOP	school= PHYSIC-LOC(ARGUMENT)	be located-PRES
	'He is located in the school.'		

which can both be represented as IN(A, B), is the **total negation** indicated in (40) and (41):

(40)	sore=wa	gakkō=ni	(*ara-)	**nai**-0
	it(SUBJ)=TOP	school=PHYSIC-LOC(ARGUMENT)	be located	**NEG**-PRES
	'It is **not** located in the school.'			

(41)	kare=wa	gakkō=ni	i-	**nai**-0
	he(SUBJ)=TOP	school= PHYSIC-LOC(ARGUMENT)	be located	**NEG**-PRES
	'He is **not** located in the school.'			

Unlike the existence-asserting and domain-specifying conjunctive E(A) ∧ IN(A, B), the location-specifying-only IN(A, B) is an atomic proposition and, naturally, its negation constitutes ¬IN(A, B), which corresponds to (40) and (41).

On the other hand, when the PP *gakkō=ni* in (40) and (41) is marked by *=wa*, this particle functions contrastively, as in (42) and (43), respectively, where CONTR abbreviates contrastive:

(42)	sore=wa	gakkō=ni=**wa**	(*ara-)	**nai**-0
	it(SUBJ)=TOP	school=PHYSIC-LOC(ARGUMENT)=**CONTR**	be located	**NEG**-PRES
	'It is **not** located in the school **but somewhere else**.'			

(43)	kare=wa	gakkō=ni=**wa**	i-	**nai**-0
	he(SUBJ)=TOP	school= PHYSIC-LOC(ARGUMENT)=**CONTR**	be located	**NEG**-PRES
	'He is **not** located in the school **but somewhere else**.'			

In contrast to (40) and (41), which are representable as ¬IN(A, B), (42) and (43) can be represented schematically as ¬IN(A, B) ∧ ∃x (IN(A, x) ∧ x ≠ B) — semantically, 'A is not located in B, and there exists a physical domain x distinct from B such that A is located in x'.

The difference in default negation pattern between the existence-asserting and domain-specifying construction and the location-specifying-only one lies in the fact that the former, **E(A) ∧ IN(A, B), is conjunctive**, whereas the latter, **IN(A, B), is atomic**. This ultimately derives from **the status of B as an adjunct in the former and as an argument in the latter**, since if B were an argument, E(A) ∧ IN(A, B) would correspond to something like EXIST-MANNER(A, B), whose negation would be the total ¬EM(A, B).

Finally, if =*de aru* were a single copular unit, partial negation with =*wa* would necessarily take the form **kare*=*wa* <u>*gakusei*=***wa***</u> *de* ***nai***, as in (44):

(44)	*kare=wa	gakusei=**wa**	de	**nai**-0
	he(SUBJ)=TOP	student(PRED)=**CONTR**	COP	**NEG**-PRES
	'He is **not** a student **but something else**.'			

The fact that this is ungrammatical further supports the separability of =*de* and *aru* and the constituency of *gakusei*=*de*.

8. The differences between *A*=*wa B*=*de aru* and *A*=*ga B*=*de aru*

The following table (45) provides a preliminary overview of the propositional identity and the information-structural differences between sentences of the type ***A*=*wa B*=*de aru*** and those of the type ***A*=*ga B*=*de aru***, without attempting a full formalisation, which lies beyond the scope of this paper:

(45)

Sentence	**Propositional structure**	**Informational structure**	
		Given	**New**
A=***wa*** *B*=*de aru*	E(A) ∧ IN(A, B)	E(A)	IN(A, B)
A=***wa*** *B*=*de*=*wa nai*	E(A) ∧ ¬IN(A, B)	E(A)	¬IN(A, B)
A=***wa*** *B*=*de*=*wa naku* *C*=***wa*** *B*=*de aru*	E(A) ∧ ¬IN(A, B) ∧ E(C) ∧ IN(C, B)	E(A) ∧ E(C)	¬IN(A, B) ∧ IN(C, B)
A=***ga*** *B*=*de aru*	E(A) ∧ IN(A, B)	∃x (E(x) ∧ IN(x, B))	x = A
A=***ga*** *B*=*de*=*wa nai*	E(A) ∧ ¬IN(A, B)	∃x (E(x) ∧ IN(x, B))	x ≠ A
A=***ga*** *B*=*de*=*wa naku* *C*=***ga*** *B*=*de aru*	E(A) ∧ ¬IN(A, B) ∧ E(C) ∧ IN(C, B)	∃x (E(x) ∧ IN(x, B))	(x ≠ A) ∧ (x = C)

Syntactically, the structure of *A*=***ga*** *B*=*de aru* in (45) can be represented as in (46):

(46) $[_{TP}\ \boxed{[_{NP}\ A{=}ga_{i}]}\ [_{T'}\ \boxed{[_{VP}\ [_{PP}\ B{=}de]\ [_{VP}\ [_{NP}\ x_{i}]\ [_{V}\ ar\text{-}]]]}\ [_{T}\ \text{-}ru]]]$

where *x* represents a trace, used here instead of the conventional *t*, and $[_{\text{V}}\ \textit{ar-}]$ has not been moved for reasons of visibility. In this structure, the VP $[_{\text{VP}}\ [_{\text{PP}}\ B{=}\textit{de}]\ [_{\text{VP}}\ [_{\text{NP}}\ x_i]\ [_{\text{V}}\ \textit{ar-}]]]$ denotes the given information ∃x (E(x) ∧ IN(x, B)), whereas the **untopicalised** — and thus capable of indicating a piece of new information — NP $[_{\text{NP}}\ A{=}\textit{ga}_i]$ denotes the new information x = A, through the anaphoric relation of the trace $[_{\text{NP}}\ x_i]$ to its antecedent $[_{\text{NP}}\ A{=}\textit{ga}_i]$. The constituents representing given and new information are enclosed within boxes and, furthermore, those representing new information are shaded. (The bracket notation adopted here is a descriptive representation of hierarchical constituency, and the traditional category labels are used merely as a device for visualising the structure.)

On the other hand, the structure of *A=**wa** B=de aru* in (45) can be represented as in (47):

$$(47)\ [_{\text{TopP}}\ \boxed{[A_i{=}\textit{wa}]}\ [_{\text{Top'}}\ [_{\text{TP}}\ [_{\text{NP}}\ x_i]\ [_{\text{T'}}\ [_{\text{VP}}\ \boxed{[_{\text{PP}}\ B{=}\textit{de}]}\ \boxed{[_{\text{VP}}\ [_{\text{NP}}\ x_i]\ [_{\text{V}}\ \textit{ar-}]]}]\ [_{\text{T}}\ \textit{-ru}]]]\ [_{\text{Top}}\ \textit{-0}]]]$$

The category *Topic*, abbreviated as Top in (47), is "a kind of functional category" that "can be considered to embed the IP into discourse and context" (Hasegawa 1999: 91, translated from Japanese), and is assumed in this paper to be realised by the phonologically null morpheme *-0*. The element that moves to the specifier of TopP is assigned the particle *=wa* and functions to indicate the topic of the sentence. In this structure, the **topicalised** phrase $[A_i{=}\textit{wa}]$ — which typically indicates a piece of given information — and the VP $[_{\text{VP}}\ [_{\text{NP}}\ x_i]\ [_{\text{V}}\ \textit{ar-}]]$ — where the verb $[_{\text{V}}\ \textit{ar-}]$ 'to exist' provides another piece of given information, since the topicalised phrase $[A_i{=}\textit{wa}]$ generally presupposes the existence of A — together denote the given information E(A). By contrast, the PP $[_{\text{PP}}\ B{=}\textit{de}]$ denotes the new information IN(A, B), since a sentence conveying only given information would be informationally vacuous. Note that, since IN(x, B) ⇒ E(x), ∃x (E(x) ∧ IN(x, B)) ≡ ∃x IN(x, B); nevertheless, the logically redundant ∃x (E(x) ∧ IN(x, B)) represents a component within the informational structure that corresponds to the syntactic structure.

9. The contracted form *=da*

In spoken language, the contracted form *=da* is often used instead of *=de aru*, which is more formal and literal, as in *kare=wa gakusei=**da*** 'he is a student'. It should be noted that this **contracted form *=da*** does not represent a new copular category but a **purely phonological**

reduction of the postpositional-verbal complex =*de aru*, as shown in (48):

(48) kare=wa gakusei=**d** **a**-0
he(SUBJ)=TOP student=**CONCEP-LOC/CONJ**(ADJUNCT) **exist**-PRES
'He exists, and it is as a student.'

The historical change from =*de aru* to =*da*, possibly via =*dea*[3], involves no syntactic or semantic reanalysis, but a process of phonetic contraction comparable to French *à le* → *au* 'to the' or Portuguese *de a* → *da* 'from the, of the'. Just as there is no semantic necessity for *à* and *le* to fuse into *au*, or for *de* and *a* into *da* — a phenomenon comparable to the English *to the* or *from the* forming a single word — there is likewise no semantic motivation for =*de* and *aru* to form a single lexical item.

That said, however, the reduction of *aru* to *a* can be accounted for in pragmatic and informational terms. Since the topicalised phrase *A=wa* generally presupposes the existence of A, and since the occurrence of *B=de* entails the existence of A by virtue of the conditional IN(A, B) ⇒ E(A), the verb *aru* becomes informationally redundant when it once again refers to *A*'s existence. Furthermore, in a sentence in which IN(A, B) is presented as new information, E(A) is presupposed and thus treated as given, owing to the same conditional IN(A, B) ⇒ E(A). In this context, E(A) is pragmatically and informationally less significant than the focus constituent IN(A, B). It may therefore be assumed that **this low pragmatic and informational salience is reflected in the phonological reduction of *aru* to *a***.

The two elements remain distinct in their underlying structure: =*de* functions as a postposition indicating the domain in which existence holds, and the stem of *aru* as the existential verb itself. This is precisely why the following highly colloquial dialogue in (49) is possible:

(49) *Kare=tte gakusei=**da** yone? — Iya, gakusei=tte yū=ka, gakusei=**de=mo** **aru**=tte kanzi kana.*

'He's a student, right?' — 'Well, if you can call him a student, I'd say he's *also* a student, kind of.'

In this dialogue, the interlocutor, recognising that the underlying structure of *gakusei=da* is *gakusei=de aru*, when asked to confirm that *kare*'s domain of existence can be identified with

[3] Historically, the form =*dea*, which yielded the modern =*da*, can be attested in Azuchi–Momoyama-period materials: *bonnin yorimo giŭzaini fuxôzuru coto***dea** "(it) is a reason for imposing a heavier guilt (on him) than on an ordinary person" (Amakusa Edition of *Esopo no fabulas*, 1593, p.475, ll.10-11, in Text Data Sets for Research on the History of Japanese).

that of 'student', replies that his existence extends to other domains as well, expressed by the form *gakusei=de=**mo** aru*, representable as E(he) ∧ IN(he, student) ∧ IN(he, something else)….

Returning to Section 1, *kare=wa gakusei=de, kanozyo=wa syakaizin=d**a*** 'he (is) a student, and she is a working adult' in (5) is thus, underlyingly, a compound sentence *kare=wa gakusei=de **ari**, kanozyo=wa syakaizin=de **aru***, in which the first predicate *ari* constitutes presupposed given information and is therefore judged to be less informationally salient, and consequently deleted as recoverable, based on the structural parallelism between the two coordinate clauses, as indicated in (50):

(50)	kare=wa	gakusei=de	(**ari**)	kanozyo=wa	syakaizin=d	**a**-0
	he(SUBJ)=TOP	student=LOC(ADJC)	(**exist**)	she(SUBJ)=TOP	working adult=LOC(ADJC)	**exist**-PRES

'He (**exist**s) as a student, she **exist**s as a working adult.'

(50) is an instance of zeugma, so *=de* is not an inflected form of *=da*. Therefore, *=da* as a whole should not be regarded as a copula in the strict sense, but as a **phonologically reduced surface realisation of the underlying construction *=de aru*** — more theoretically, ***=de ar-0***, where *ar-* and *-0* represent the stem of the existential verb *aru* and a phonologically zero tense suffix specified as [−past].

10. Conclusion — *=da* not as a copula but as the conceptual-locative postposition *=de* followed by the one-argument existential verb *aru*

In summary, Japanese has no genuine copula — *da* is not *be*. The construction *A=wa B dearu* 'A exists, and its domain of existence is B' consists of a **one-argument, existence-asserting core construction [*A=wa aru*] 'A exists'** and a **domain-specifying adjunct [*B=de*], meaning 'and in the conceptual domain denoted by B' or, more symply, 'and as B'**. It should therefore be syntactically represented as ***A=wa B=de aru***. Consequently, treating *dearu* or its contracted form *da* as a single copula is structurally problematic. The form *dearu* is more appropriately analysed as an **existence-and-domain complex**, consisting of the one-argument existential verb *aru* and the conceptual-locative and conjunctive postposition *=de*, which conjunctively specifies the domain in which the existence of the theme is realised. The more colloquial form *=da* is simply a phonologically contracted form corresponding to the more formal and literal *=de aru*.

Thus, through the above syntactic and semantic analysis, it has been possible to account, by means of a single, consistent principle — namely, the **assertion of existence by [NP_1=*wa* + *ar-*]** and the **conjunctive specification of the domain of existence by [NP_2 + =*de*]**, schematically representable as **EXIST(NP_1) ∧ IN(NP_1, NP_2), not reducible to BE(NP_1, NP_2)** — for:

- Morphosyntactic separability of *de* and *aru* in *dearu*
- Optional omission of *aru* in *dearu*
- Prosodic-word boundary between *de* and *aru*
- Syntactic and prosodic constituency of *de* and the NPs preceding it
- Grounds for analysing =*de* as an adjunct-marking postposition
- Foregrounding of the lexical meaning 'to exist' of the verb *aru*
- Grounds on which the Japanese desiderative =*de ari-tai* is not equivalent to the English *to want to be*
- Diachronic split between *A B=ni ari* → *A B=nite ari* → *A=wa B=de aru* and *A B=ni ari* → *A=wa B=ni aru/iru*
- Grounds on which the default negation of *A=wa B=de aru* is not the total *A=wa B=de nai* but the partial *A=wa B=de=wa nai*
- Syntactic and information-structural differences between *A=wa B=de aru* and *A=ga B=de aru*
- Underlying structure =*de ar-0* surfacing as =*da*
- Pragmatic and informational motivation for the phonological reduction *aru* → *a* in =*da*

Finally, while the present study reinterprets =*da* and =*de aru* as a postpositional-verbal complex expressing E(A) ∧ IN(A, B) rather than a copular predicate denoting BE(A, B), this proposal is not intended to invalidate previous research conducted within the copular framework. Much of the empirical and descriptive work on so-called copular sentences remains valuable and can, in fact, be accommodated within the present account as analyses of the IN(A, B) component — that is, as investigations into how the domain of existence is structured. In this respect, the present analysis may be regarded not as a rejection of the copular tradition, but as an attempt to extend it by situating its insights within a broader model of Japanese predicational structure. Furthermore, the interaction between the existence-asserting component E(A) and the domain-specifying component IN(A, B), which has received relatively little explicit attention in previous research, may offer a possible perspective for future studies on how these two semantic components relate to one another in information structure, negation and discourse.

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